

Database Schema

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Tables	12
Foreign key relationships	17
Indexes	14
Triggers	3
Enums	2
Extensions	vector, pgcrypto

Enums

Name	Values
document_status	uploading, processing, ready, failed
message_role	system, user, assistant

Relationships

From	To	Rule
user_sessions.user_id	users.id	user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
conversations.user_id	users.id	user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
messages.conversation_id	conversations.id	conversation_id UUID NOT NULL REFERENCES conversations(id) ON DELETE CASCADE
messages.parent_message_id	messages.id	parent_message_id UUID REFERENCES messages(id) ON DELETE SET NULL
user_documents.user_id	users.id	user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
user_documents.document_id	documents.id	document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE
conversation_documents.conversation_id	conversations.id	conversation_id UUID NOT NULL REFERENCES conversations(id) ON DELETE CASCADE
conversation_documents.document_id	documents.id	document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE
attachments.message_id	messages.id	message_id UUID NOT NULL REFERENCES messages(id) ON DELETE CASCADE
attachments.document_id	documents.id	document_id UUID REFERENCES documents(id) ON DELETE SET NULL
document_chunks.document_id	documents.id	document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE

From	To	Rule
chunk_embeddings.chunk_id	document_chunks.id	chunk_id UUID NOT NULL REFERENCES document_chunks(id) ON DELETE CASCADE
chunk_embeddings.embedding_model_id	embedding_models.id	embedding_model_id SMALLINT NOT NULL REFERENCES embedding_models(id)
message_citations.message_id	messages.id	message_id UUID NOT NULL REFERENCES messages(id) ON DELETE CASCADE
message_citations.chunk_id	document_chunks.id	chunk_id UUID NOT NULL REFERENCES document_chunks(id) ON DELETE CASCADE
message_citations.embedding_model_id	embedding_models.id	embedding_model_id SMALLINT NOT NULL REFERENCES embedding_models(id)
message_citations.chunk_id, embedding_model_id	chunk_embeddings.chunk_id, embedding_model_id	FOREIGN KEY (chunk_id, embedding_model_id) REFERENCES chunk_embeddings(chunk_id, embedding_m...

Table: users

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
username	VARCHAR(50)	NOT NULL
email	VARCHAR(255)	NOT NULL
password_hash	TEXT	NOT NULL
profile_picture	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Index	Definition
UNIQUE idx_users_email_lower	users (LOWER(email))
UNIQUE idx_users_username_lower	users (LOWER(username))

Table: user_sessions

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE
refresh_token	TEXT	NOT NULL
device	TEXT	-
ip_address	INET	-
is_revoked	BOOLEAN	NOT NULL DEFAULT false
expires_at	TIMESTAMPTZ	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Table: embedding_models

Column	Type	Details
id	SMALLSERIAL	PRIMARY KEY
name	VARCHAR(100)	UNIQUE NOT NULL
--	e.g. 'text-embedding-3-large' dimension SMALLINT	NOT NULL
--	e.g. 1536 created_at TIMESTAMPTZ	NOT NULL DEFAULT now()

Table: conversations

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE
title	VARCHAR(255)	NOT NULL DEFAULT 'New conversation'
model_name	VARCHAR(100)	NOT NULL
system_prompt	TEXT	-
generation_config	JSONB	NOT NULL DEFAULT '{}::jsonb
is_archived	BOOLEAN	NOT NULL DEFAULT false
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Table: messages

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
conversation_id	UUID	NOT NULL REFERENCES conversations(id) ON DELETE CASCADE
parent_message_id	UUID	REFERENCES messages(id) ON DELETE SET NULL
role	message_role	NOT NULL
content	TEXT	NOT NULL
model_name	VARCHAR(100)	-
token_count	INTEGER	-
generation_time	REAL	-
is_helpful	BOOLEAN	-
feedback_text	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Table: documents

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
name	VARCHAR(255)	NOT NULL
description	TEXT	-
storage_path	TEXT	NOT NULL
content_hash	CHAR(64)	NOT NULL
mime_type	VARCHAR(100)	NOT NULL

Column	Type	Details
file_size	BIGINT	NOT NULL
status	document_status	NOT NULL DEFAULT 'uploading'
uploaded_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Table: user_documents

Column	Type	Details
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE
document_id	UUID	NOT NULL REFERENCES documents(id) ON DELETE CASCADE
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Table constraints
PRIMARY KEY (user_id, document_id)

Table: conversation_documents

Column	Type	Details
conversation_id	UUID	NOT NULL REFERENCES conversations(id) ON DELETE CASCADE
document_id	UUID	NOT NULL REFERENCES documents(id) ON DELETE CASCADE
added_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Table constraints
PRIMARY KEY (conversation_id, document_id)

Table: attachments

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
message_id	UUID	NOT NULL REFERENCES messages(id) ON DELETE CASCADE
type	VARCHAR(50)	NOT NULL
original_name	TEXT	NOT NULL
stored_name	TEXT	NOT NULL
mime_type	VARCHAR(100)	NOT NULL
file_size	BIGINT	NOT NULL
storage_path	TEXT	NOT NULL
thumbnail_path	TEXT	-
document_id	UUID	REFERENCES documents(id) ON DELETE SET NULL

Column	Type	Details
created_at	TIMESTAMPZ	NOT NULL DEFAULT now()

Table: document_chunks

Column	Type	Details
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()
document_id	UUID	NOT NULL REFERENCES documents(id) ON DELETE CASCADE
chunk_number	INTEGER	NOT NULL
page_number	INTEGER	-
chunk_text	TEXT	NOT NULL
token_count	INTEGER	-
created_at	TIMESTAMPZ	NOT NULL DEFAULT now()

Table constraints
UNIQUE (document_id, chunk_number)

Table: chunk_embeddings

Column	Type	Details
chunk_id	UUID	NOT NULL REFERENCES document_chunks(id) ON DELETE CASCADE
embedding_model_id	SMALLINT	NOT NULL REFERENCES embedding_models(id)
embedding	VECTOR(1536)	NOT NULL
created_at	TIMESTAMPZ	NOT NULL DEFAULT now()

Table constraints
PRIMARY KEY (chunk_id, embedding_model_id)

Index	Definition
idx_chunk_embeddings_hnsw	chunk_embeddings USING hnsw (embedding vector_cosine_ops)

Table: message_citations

Column	Type	Details
message_id	UUID	NOT NULL REFERENCES messages(id) ON DELETE CASCADE
chunk_id	UUID	NOT NULL REFERENCES document_chunks(id) ON DELETE CASCADE
embedding_model_id	SMALLINT	NOT NULL REFERENCES embedding_models(id)
similarity_score	REAL	-

Column	Type	Details
rank	SMALLINT	-
Table constraints		
PRIMARY KEY (message_id, chunk_id, embedding_model_id)		
FOREIGN KEY (chunk_id, embedding_model_id) REFERENCES chunk_embeddings(chunk_id, embedding_model_id) ON DELETE CASCADE		

Triggers

Name	Event	Function
update_users_updated_at	BEFORE UPDATE ON users FOR EACH ROW	update_updated_at_column()
update_user_sessions_updated_at	BEFORE UPDATE ON user_sessions FOR EACH ROW	update_updated_at_column()
update_conversations_updated_at	BEFORE UPDATE ON conversations FOR EACH ROW	update_updated_at_column()

Source SQL

```
-- =====
-- EXTENSIONS
-- =====
CREATE EXTENSION IF NOT EXISTS vector;
CREATE EXTENSION IF NOT EXISTS pgcrypto;

-- =====
-- ENUMS
-- =====
CREATE TYPE document_status AS ENUM ('uploading', 'processing', 'ready', 'failed');
CREATE TYPE message_role AS ENUM ('system', 'user', 'assistant');

-- =====
-- AUTOMATED TIMESTAMP TRIGGER
-- =====
CREATE OR REPLACE FUNCTION update_updated_at_column()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = now();
    RETURN NEW;
END;
$$ language 'plpgsql';

-- =====
-- 1. USERS
-- =====
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    username VARCHAR(50) NOT NULL,
    email VARCHAR(255) NOT NULL,
    password_hash TEXT NOT NULL,
    profile_picture TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE UNIQUE INDEX idx_users_email_lower ON users (LOWER(email));
CREATE UNIQUE INDEX idx_users_username_lower ON users (LOWER(username));

CREATE TRIGGER update_users_updated_at
BEFORE UPDATE ON users
FOR EACH ROW
EXECUTE FUNCTION update_updated_at_column();

-- =====
-- 2. USER SESSIONS
-- =====
CREATE TABLE user_sessions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    refresh_token TEXT NOT NULL,
    device TEXT,
    ip_address INET,
    is_revoked BOOLEAN NOT NULL DEFAULT false,
    expires_at TIMESTAMPTZ NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

```

);
CREATE INDEX idx_sessions_user ON user_sessions(user_id);
CREATE UNIQUE INDEX idx_sessions_token ON user_sessions(refresh_token);

CREATE TRIGGER update_user_sessions_updated_at
    BEFORE UPDATE ON user_sessions
    FOR EACH ROW
    EXECUTE FUNCTION update_updated_at_column();

-- =====
-- 3. EMBEDDING MODELS
-- =====
-- Registry of models available to the system. Note: chunk_embeddings.embedding
-- is currently pinned to VECTOR(1536) – registering a model with a different
-- dimension will fail on insert, not at registration time. See table 11 below.
CREATE TABLE embedding_models (
    id SMALLSERIAL PRIMARY KEY,
    name VARCHAR(100) UNIQUE NOT NULL,      -- e.g. 'text-embedding-3-large'
    dimension SMALLINT NOT NULL,           -- e.g. 1536
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

-- =====
-- 4. CONVERSATIONS
-- =====
CREATE TABLE conversations (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    title VARCHAR(255) NOT NULL DEFAULT 'New conversation',
    model_name VARCHAR(100) NOT NULL,
    system_prompt TEXT,
    generation_config JSONB NOT NULL DEFAULT '{}::jsonb',
    is_archived BOOLEAN NOT NULL DEFAULT false,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_conversations_user ON conversations(user_id);

CREATE TRIGGER update_conversations_updated_at
    BEFORE UPDATE ON conversations
    FOR EACH ROW
    EXECUTE FUNCTION update_updated_at_column();

-- =====
-- 5. MESSAGES
-- =====
CREATE TABLE messages (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    conversation_id UUID NOT NULL REFERENCES conversations(id) ON DELETE CASCADE,
    parent_message_id UUID REFERENCES messages(id) ON DELETE SET NULL,
    role message_role NOT NULL,
    content TEXT NOT NULL,
    model_name VARCHAR(100),
    token_count INTEGER,
    generation_time REAL,
    is_helpful BOOLEAN,
    feedback_text TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_messages_conv_created ON messages(conversation_id, created_at);

-- =====
-- 6. DOCUMENTS
-- =====
CREATE TABLE documents (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name VARCHAR(255) NOT NULL,
    description TEXT,
    storage_path TEXT NOT NULL,
    content_hash CHAR(64) NOT NULL,
    mime_type VARCHAR(100) NOT NULL,
    file_size BIGINT NOT NULL,
    status document_status NOT NULL DEFAULT 'uploading',
    uploaded_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE UNIQUE INDEX idx_documents_hash ON documents(content_hash);

-- =====

```

```

-- 7. USER DOCUMENTS
-- =====
CREATE TABLE user_documents (
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    PRIMARY KEY (user_id, document_id)
);
CREATE INDEX idx_user_documents_doc ON user_documents(document_id);

-- =====
-- 8. CONVERSATION DOCUMENTS
-- =====
CREATE TABLE conversation_documents (
    conversation_id UUID NOT NULL REFERENCES conversations(id) ON DELETE CASCADE,
    document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
    added_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    PRIMARY KEY (conversation_id, document_id)
);
CREATE INDEX idx_convdocs_document ON conversation_documents(document_id);

-- =====
-- 9. ATTACHMENTS
-- =====
CREATE TABLE attachments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    message_id UUID NOT NULL REFERENCES messages(id) ON DELETE CASCADE,
    type VARCHAR(50) NOT NULL,
    original_name TEXT NOT NULL,
    stored_name TEXT NOT NULL,
    mime_type VARCHAR(100) NOT NULL,
    file_size BIGINT NOT NULL,
    storage_path TEXT NOT NULL,
    thumbnail_path TEXT,
    document_id UUID REFERENCES documents(id) ON DELETE SET NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_attachments_message ON attachments(message_id);

-- =====
-- 10. DOCUMENT CHUNKS (text, independent of embedding model)
-- =====
CREATE TABLE document_chunks (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
    chunk_number INTEGER NOT NULL,
    page_number INTEGER,
    chunk_text TEXT NOT NULL,
    token_count INTEGER,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    UNIQUE (document_id, chunk_number)
);
CREATE INDEX idx_chunks_document ON document_chunks(document_id);

-- =====
-- 11. CHUNK EMBEDDINGS
-- =====
-- Pinned to VECTOR(1536). If you register a model with a different dimension
-- in embedding_models, inserts for that model will fail here – either keep
-- the system single-dimension, or split this table per dimension when you
-- actually need a second model.
CREATE TABLE chunk_embeddings (
    chunk_id UUID NOT NULL REFERENCES document_chunks(id) ON DELETE CASCADE,
    embedding_model_id SMALLINT NOT NULL REFERENCES embedding_models(id),
    embedding VECTOR(1536) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    PRIMARY KEY (chunk_id, embedding_model_id)
);

CREATE INDEX idx_chunk_embeddings_hnsw
    ON chunk_embeddings USING hnsw (embedding vector_cosine_ops);

-- Supports filtering by model before/alongside the ANN search
CREATE INDEX idx_chunk_embeddings_model ON chunk_embeddings(embedding_model_id);

-- =====
-- 12. MESSAGE CITATIONS
-- =====

```

```

-- embedding_model_id records which model's embedding actually produced the
-- similarity_score/rank for this citation. Composite FK ties it back to the
-- specific (chunk_id, embedding_model_id) row in chunk_embeddings, so a
-- citation can never reference a model/chunk pair that was never embedded.
CREATE TABLE message_citations (
    message_id UUID NOT NULL REFERENCES messages(id) ON DELETE CASCADE,
    chunk_id UUID NOT NULL REFERENCES document_chunks(id) ON DELETE CASCADE,
    embedding_model_id SMALLINT NOT NULL REFERENCES embedding_models(id),
    similarity_score REAL,
    rank SMALLINT,
    PRIMARY KEY (message_id, chunk_id, embedding_model_id),
    FOREIGN KEY (chunk_id, embedding_model_id)
        REFERENCES chunk_embeddings(chunk_id, embedding_model_id)
        ON DELETE CASCADE
);
CREATE INDEX idx_citations_chunk ON message_citations(chunk_id);

-- =====
-- Example similarity search
-- =====
-- SELECT dc.chunk_text, dc.page_number, ce.chunk_id
-- FROM chunk_embeddings ce
-- JOIN document_chunks dc ON dc.id = ce.chunk_id
-- JOIN conversation_documents cd ON cd.document_id = dc.document_id
-- WHERE cd.conversation_id = $1
-- AND ce.embedding_model_id = $2
-- ORDER BY ce.embedding <=> $3
-- LIMIT 5;

```